Операционные системы

Поиск файлов. Перенаправление ввода-вывода. Просмотр запущенных процессов

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1. Цели и задачи работы

Цель лабораторной работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

Задачи лабораторной работы

1 Изучить перенаправление ввода-вывода

2 Изучить работу фильтров

3 Изучить команду поиска

4 Ознакомиться с управлением процессами

5 Ознакомиться с командами df du

2. Процесс выполнения лабораторной работы

Перенаправление ввода-вывода

```
jamalsaidmagomedov@jamalsaidmagomedov:~$ ls /etc/ > file.txt
jamalsaidmagomedov@jamalsaidmagomedov:~$ ls >> file.txt
jamalsaidmagomedov@jamalsaidmagomedov:~$ cat file.txt
abrt
adjtime
aliases
alsa
alternatives
anaconda
anacrontab
anthy-unicode.conf
asound.conf
at.deny
audit
authselect
avahi
bash_completion.d
bashrc
bindresvport.blacklist
binfmt.d
bluetooth
brlapi.key
brltty
brltty.conf
```

Рисунок 1: Запись в файл

Работа фильтра

```
jamalsaidmagomedov@jamalsaidmagomedov:~$ grep .conf file.txt > conf.txt
jamalsaidmagomedov@jamalsaidmagomedov:~$ cat conf.txt
anthy-unicode.conf
asound.conf
brltty.conf
chrony.conf
cpupower-service.conf
dconf
dnsmasq.conf
dracut.conf
dracut.conf.d
fprintd.conf
fuse.conf
host.conf
idmapd.conf
kdump.conf
krb5.conf
krb5.conf.d
ld.so.conf
ld.so.conf.d
libaudit.conf
libuser.conf
locale.conf
```

Рисунок 2: Поиск расширения .conf

Команда поиска

```
/home/jamalsaidmagomedov/.npm/_cacache/index-v5/ce
/home/jamalsaidmagomedov/.npm/_cacache/index-v5/84/ce
/home/jamalsaidmagomedov/.npm/_cacache/index-v5/09/c0
/home/jamalsaidmagomedov/.npm/_cacache/index-v5/09/c0/cce94e2c664a8f390a8f1ade3346d5c95f3ced5cf6fc29def30
220242c53
/home/jamalsaidmagomedov/.npm/_cacache/index-v5/59/e7/cb225f160e685933e8439d3cf624ef721a58996cfad55fb2000
f59e2edcc
/home/jamalsaidmagomedov/.npm/_cacache/index-v5/cd
/home/jamalsaidmagomedov/.npm/_cacache/index-v5/c9
/home/jamalsaidmagomedov/.npm/_cacache/index-v5/c0
/home/jamalsaidmagomedov/.npm/_cacache/index-v5/44/44/c9a2c5dc2999cbe868c370ef479e17dfd1e86117d1350482a1d
990504757
/home/jamalsaidmagomedov/.npm/_cacache/index-v5/9f/ae/c077854801f7b4b5cf7eea2bb68ca2bc53f0b53cea310cda0b2
03217f693
/home/jamalsaidmagomedov/.npm/_cacache/index-v5/af/cf
/home/jamalsaidmagomedov/.npm/_cacache/index-v5/c3
/home/jamalsaidmagomedov/.npm/_cacache/index-v5/78/34/c1e131014ad3a4ee13f096e70f74c03e68837bc2c8188a44d55
4f3f3ba2e
/home/jamalsaidmagomedov/.npm/_cacache/index-v5/a1/5e/c9aa1c46bf0c67511379e97bd07fca87532422f2506214f94db
8861160d3
/home/jamalsaidmagomedov/.npm/_cacache/index-v5/ca
/home/jamalsaidmagomedov/snap/hugo/common
/home/jamalsaidmagomedov/snap/hugo/current
/home/jamalsaidmagomedov/conf.txt
jamalsaidmagomedov@jamalsaidmagomedov:~$
```

Рисунок 3: Поиск файлов

Команда поиска

```
find: '/etc/lvm/backup': Permission denied
find: '/etc/lvm/cache': Permission denied
find: '/etc/lvm/devices': Permission denied
find: '/etc/nftables': Permission denied
find: '/etc/openvpn/client': Permission denied
find: '/etc/openvpn/server': Permission denied
find: '/etc/pki/rsyslog': Permission denied
/etc/hp
/etc/hp/hplip.conf
/etc/httpd
/etc/httpd/conf/httpd.conf
/etc/libibverbs.d/hfi1verbs.driver
/etc/libibverbs.d/hns.driver
/etc/logrotate.d/httpd
/etc/nvme/hostnqn
/etc/nvme/hostid
find: '/etc/polkit-1/localauthority': Permission denied
find: '/etc/polkit-1/rules.d': Permission denied
find: '/etc/sos/cleaner': Permission denied
/etc/sane.d/dll.d/hpaio
/etc/sane.d/hp.conf
/etc/sane.d/hp3900.conf
/etc/sane.d/hp4200.conf
/etc/sane.d/hp5400.conf
/etc/sane.d/hpsj5s.conf
:find: '/etc/ssh/sshd_config.d': Permission denied
find: '/etc/sssd': Permission denied
find: '/etc/sudoers.d': Permission denied
find: '/etc/userdb': Permission denied
```

Рисунок 4: Поиск файлов

Управление процессами

```
jamalsaidmagomedov@jamalsaidmagomedov:~$
jamalsaidmagomedov@jamalsaidmagomedov:~$ find ~ -name "log*" > logfile &
[1] 6558
jamalsaidmagomedov@jamalsaidmagomedov:~$
[1]+  Done                    find ~ -name "log*" > logfile
jamalsaidmagomedov@jamalsaidmagomedov:~$ rm logfile
jamalsaidmagomedov@jamalsaidmagomedov:~$
```

Рисунок 5: Фоновый запуск процесса

Управление процессами

```
jamalsaidmagomedov@jamalsaidmagomedov:~$
jamalsaidmagomedov@jamalsaidmagomedov:~$ gedit &
[1] 6596
jamalsaidmagomedov@jamalsaidmagomedov:~$ ps | grep gedit
   6596 pts/0    00:00:00 gedit
jamalsaidmagomedov@jamalsaidmagomedov:~$ kill 6596
[1]+  Terminated              gedit
jamalsaidmagomedov@jamalsaidmagomedov:~$
jamalsaidmagomedov@jamalsaidmagomedov:~$
```

Рисунок 6: Фоновый запуск и завершение процесса

Команды df и du

```
DF(1)                                User Commands                                DF(1)

NAME
       df - report file system space usage

SYNOPSIS
       df [OPTION]... [FILE]...

DESCRIPTION
       This  manual  page documents the GNU version of df.  df displays the amount of space available on
       the file system containing each file name argument.  If no file name is given, the  space  avail-
       able on all currently mounted file systems is shown.  Space is shown in 1K blocks by default, un-
       less the environment variable POSIXLY_CORRECT is set, in which case 512-byte blocks are used.

       If  an  argument  is the absolute file name of a device node containing a mounted file system, df
       shows the space available on that file system rather than on the file system containing  the  de-
       vice node.  This version of df cannot show the space available on unmounted file systems, because
       on  most kinds of systems doing so requires non-portable intimate knowledge of file system struc-
       tures.

OPTIONS
       Show information about the file system on which each FILE resides, or all  file  systems  by  de-
       fault.

       Mandatory arguments to long options are mandatory for short options too.

       -a, --all
              include pseudo, duplicate, inaccessible file systems

       -B, --block-size=SIZE
              scale  sizes  by SIZE before printing them; e.g., '-BM' prints sizes in units of 1,048,576
              bytes; see SIZE format below
 Manual page df(1) line 1 (press h for help or q to quit)
```

Рисунок 7: Справка по команде df

Команды df и du

```
DU(1)                                User Commands                                DU(1)

NAME
       du - estimate file space usage

SYNOPSIS
       du [OPTION]... [FILE]...
       du [OPTION]... --files0-from=F

DESCRIPTION
       Summarize device usage of the set of FILEs, recursively for directories.

       Mandatory arguments to long options are mandatory for short options too.

       -0, --null
              end each output line with NUL, not newline

       -a, --all
              write counts for all files, not just directories

       --apparent-size
              print  apparent  sizes  rather  than  device  usage; although the apparent size is usually
              smaller, it may be larger due to holes in ('sparse') files, internal fragmentation,  indi-
              rect blocks, and the like

       -B, --block-size=SIZE
              scale  sizes  by SIZE before printing them; e.g., '-BM' prints sizes in units of 1,048,576
              bytes; see SIZE format below

       -b, --bytes
              equivalent to '--apparent-size --block-size=1'

 Manual page du(1) line 1 (press h for help or q to quit)
```

Рисунок 8: Запуск команды df

Команды df и du

```
jamalsaidmagomedov@jamalsaidmagomedov:~$
jamalsaidmagomedov@jamalsaidmagomedov:~$ df
Filesystem     1K-blocks     Used Available Use% Mounted on
/dev/nvme0n1p3 155186176 91430980  63165804  60% /
devtmpfs         4011288        0   4011288   0% /dev
tmpfs            4035664      100   4035564   1% /dev/shm
tmpfs            1614268     1968   1612300   1% /run
tmpfs               1024        0      1024   0% /run/credentials/systemd-journald.service
tmpfs            4035664     5724   4029940   1% /tmp
/dev/nvme0n1p3 155186176 91430980  63165804  60% /home
/dev/nvme0n1p2   1992552   518800   1352512  28% /boot
/dev/loop2         49280    49280         0 100% /var/lib/snapd/snap/snapd/25935
/dev/loop0         68480    68480         0 100% /var/lib/snapd/snap/core24/1349
/dev/loop1        115712   115712         0 100% /var/lib/snapd/snap/hugo/25676
tmpfs               1024        0      1024   0% /run/credentials/systemd-resolved.service
tmpfs             807132      176    806956   1% /run/user/1067
/dev/sr0         2677920  2677920         0 100% /run/media/jamalsaidmagomedov/Fedora-WS-Live-43
jamalsaidmagomedov@jamalsaidmagomedov:~$
```

Рисунок 9: Справка по команде du

Команды df и du

```
4        ./.npm/_cacache/index-v5/30/1a
4        ./.npm/_cacache/index-v5/30
424      ./.npm/_cacache/index-v5
115644   ./.npm/_cacache
115708   ./.npm
12       ./snap/hugo/25676/.config/go/telemetry/local
0        ./snap/hugo/25676/.config/go/telemetry/upload
12       ./snap/hugo/25676/.config/go/telemetry
12       ./snap/hugo/25676/.config/go
12       ./snap/hugo/25676/.config
12       ./snap/hugo/25676
0        ./snap/hugo/common
16       ./snap/hugo
16       ./snap
0        ./monthly
0        ./reports/monthly/monthly
0        ./reports/monthly
0        ./reports
4        ./ski.plases/equipment
0        ./ski.plases/plans
4        ./ski.plases
0        ./australia
0        ./play/games/play
0        ./play/games
0        ./play
1432996  .
jamalsaidmagomedov@jamalsaidmagomedov:~$
```

Рисунок 10: Запуск команды du

Команда поиска

```
/home/jamalsaidmagomedov/.npm/_cacache/index-v5/a7/81
/home/jamalsaidmagomedov/.npm/_cacache/index-v5/aa
/home/jamalsaidmagomedov/.npm/_cacache/index-v5/aa/bf
/home/jamalsaidmagomedov/.npm/_cacache/index-v5/30
/home/jamalsaidmagomedov/.npm/_cacache/index-v5/30/1a
/home/jamalsaidmagomedov/snap
/home/jamalsaidmagomedov/snap/hugo
/home/jamalsaidmagomedov/snap/hugo/25676
/home/jamalsaidmagomedov/snap/hugo/25676/.config
/home/jamalsaidmagomedov/snap/hugo/25676/.config/go
/home/jamalsaidmagomedov/snap/hugo/25676/.config/go/telemetry
/home/jamalsaidmagomedov/snap/hugo/25676/.config/go/telemetry/local
/home/jamalsaidmagomedov/snap/hugo/25676/.config/go/telemetry/upload
/home/jamalsaidmagomedov/snap/hugo/common
/home/jamalsaidmagomedov/monthly
/home/jamalsaidmagomedov/reports
/home/jamalsaidmagomedov/reports/monthly
/home/jamalsaidmagomedov/reports/monthly/monthly
/home/jamalsaidmagomedov/ski.plases
/home/jamalsaidmagomedov/ski.plases/equipment
/home/jamalsaidmagomedov/ski.plases/plans
/home/jamalsaidmagomedov/australia
/home/jamalsaidmagomedov/play
/home/jamalsaidmagomedov/play/games
/home/jamalsaidmagomedov/play/games/play
jamalsaidmagomedov@jamalsaidmagomedov:~$
```

Рисунок 11: Поиск директорий

3. Выводы по проделанной работе

Вывод

В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.